float i, E=0.05;
cin>>int n;
int cont=1;

Float sum=0;
Float z=1

Q3: Write a C++program to find the sum of following series until the fraction becomes less or equal to small constant E=0.05. : (7 marks)

$$sum = 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \cdots + \frac{1}{n}$$

$\frac{1}{cont}$

while(z > E)

sum=sum+(float) $\frac{1}{cont}$

z=sum

Q4: Write a C++ program to Swap the elements in main diagonal with the secondary diagonal in 2-D array. (7 marks)

Q5: Write a C++ program using functions to perform the following tasks: (7 marks)

- Create a function called welcome() to display a welcome message for the user of a game.
- Create another function called level(int age) to determine the player level according to the following conditions:

Age	Level
Less than 10	Beginner
10 – 18	Intermediate
Greater than 18	Advanced

Q6: Write a C++program to print the following pattern using nested loops: (answer A or B) **(7 marks)**

A.

```
*               *
*  $         $  *
*     $   $     *
*       $       *
*     $   $     *
*  $         $  *
*               *
```

B.

```
1 2 3 4 5
1 2 3 5
1 2 5
1 5
5
```

for(int cont ...

For(int i=1; i<=

Examiner | **Head of the Department** | **Dean of the Collage**

Noor Almoosawi\ Ammar Asaad | Asst. Pro. Baidaa AbdulQader | Prof. Dr. Husham Kadhum Hashim

University of Basrah
College of Computer Science and Information Technology
Computer Information System

Subject: Computer ProgrammingII
Class: 1st year
Semester: 1st Semester

Exam: Final/ 1st Attempt.
Date: 3/6/2026.
Time: 120 Minutes.

N.B.
(1) The maximum mark for this paper is 35 [7 for each question]
(2) This exam has TWO pages. Make sure you have all the pages before you begin.
(3) answer **Five** question only.

(7 marks)

Q1: A. Choose the correct answer : (choose seven only)

1. Which loop executes at least one time?
   a. While b. For c. Do .. while d. If

2. Which statement is correct for function declaration?
   a. int sum(); b. sum int(); c. function sum; d. declares sum();

3. int s1[5] = { 4}; what will be the value of s1[5] element?
   a. 4 b. unknown value c. 0 d. 1

4. Which statement skips the current iteration?
   a. break b. continue c. skip d. exit

5. *for(int i = 0 ; i<10 ; i--) , this for loop is:*
   a. *executes only one time* b. *Syntex error* c. an infinite loop d. repeat until i = 0

6. *Which function call method changes the original variable value?*
   a. *Call by value* b. call by reference c. local call d. static call

7. *What will happen if a function is called without passing the required arguments?*
   a. *the program repeats automatically* b. syntax or compilation error occurs
   c. the function returns zero d. the arguments are created automatically

8. Which statement correctly sends an array **a[5]** to a function **fun**?
   a. fun(a); b. a(fun); c. fun(a[9]); d. function[array];

Q2: Find the output: (7 marks)

1.
```
int main() {
int arr[3] = {1, 2, 3};
for(int i = 0 ; i<3 ; i++)
    arr[i] = arr[i] + arr[ 2-i];
for(int i = 0 ; i<3 ; i++)
    cout<<arr[i]<<" ";
return 0; }
```

2.
```
int fun(float a, float b){
return a+b;}

int main()
{ float a=2.5, b= 3;
  cout<<fun(a,b);}
```

3.
```
int main()
{ int n = 10;
  while(n > 0) {
    if(n % 3 == 0)
      cout << n / 2 << " ";
    n = n - 2; }
  return 0;}
```

2.5 + 3

Q3: Chosen one only. **(9 Mark)**

اقرأ　　　　　　　　　　　　　　　　اطبع

A) Design a function to read a sequence of numbers (N), then print the time's repeat of a specific number.

~~**B)** Write a C++ program to read two 1-D arrays, print them and print their sum using three functions.~~

Q4: Answer the following: (ONE ONLY) **(9 Mark)**

استبدل

1. Write a C++ program to Swap the elements in main diagonal with the secondary diagonal in 2-D array. مصفوفة ثنائية البعد
2. ~~Write a C++ program to generate one dimensional arrays to store summation of each column in 2D matrix.~~

Q5: Answer the following: **(9 Mark)**

Write a C++ program to find sum of the following series.

$$sum = \frac{x-2}{10!} - \frac{x+4}{30!} + \frac{x-6}{90!} - \cdots$$

```
#include <iostream>
using namespace std;
int main(){
float a= x-2, b=10!;
int f=1, sum=0 , n;
cout<< "Enter n: ";
cin>>n
for(int i=0; i<=n; i++)
```

Examiner	Head of the Department	The Dean
Lecturer .Noor M. Almoosawi Lecturer. Ali Abdulrazzaq	Prof. Dr. Haidar M. Al-Mashhadi	الأستاذ الدكتور Prof. Dr. Hisham K. Hashim معاون العميد للشؤون العلمية والدراسات العليا

B) What is the output of the following C++ programs? (ONLY TWO) **(4 Marks)**

(A

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++) {
      if (i % 2 == 1)
         cout << "1 ";
      else
         cout << "2 ";
   }
   cout << endl;
}
```

(B

```
int sum=0;
int a=99;
int f=1;
int n=4;
for (int i=1;i<=n;i++){
  sum=sum + f  *a;
  f*= -1;
  a -= 11;}
cout << " sum = " << sum;
```

(C

```
int n = 8;
int a [ ] = { 18, 25, 36, 44, 12, 60, 75, 89 };
int x = a [ 0 ];
for ( int i = 1; i < n; i++ )
if ( a [ i ] < x)
x = a [ i ];
cout << "The x number is: " << x;
```

(D

```
int table[3][3] = {
   {1, 2, 3},
   {4, 5, 6},
   {7, 8, 9} };
int result = 0;
int rowIndex = 2;
for (int c = 0; c < 3; c++)
   result += table[rowIndex][c];
cout << " result = " << result;
```

Q2: Answer the following: **(9 Marks)**

A) write a C++ program with a function called max(). This function receives num1 and num2 as parameters and returns the maximum value **(5 Marks)**

B) Write a suitable statement for the following (TWO ONLY): **(4 Marks)**

1. Write only function mult() that accepts two floating-point, multiplies them, and returns the result.
2. Write only a function named square() that takes an integer, computes its square, and returns the result.
3. Write only function that copies the value of the third element to the second in an array.

Final Exam of Computer Programming II
University of Basrah
College of Computer Science and Information Technology
Department of Computer Information Systems
2024-2025/ Second Semester

Exam Duration: 2 Hours | **Stage 1/ First Attempt** | **Date: 15-6-2025**

N.B.
(1) The maximum mark for this paper is **35 [8+9+9+9]**
(2) This exam has **THREE** pages. Make sure you have all the pages before you begin.
(3) Please read each question carefully before you start to answer it.
(4) <u>You have to leave one question that have (9) Marks only</u>.

Q1: A) Answer The following questions with True or False and Correct the error if False

(4 Marks)

1. This code finds the sum of the first n even numbers. sum = 0; a = 2; cout << "Enter n = "; cin >> n; for (int i = 1; i <= n; i++) { sum += a; a += 2; }	2. This program prints A in the first row, B in the second, and it continues. char ch = 'A'; int n = 5; for (int i = 1; i <= n; i++) { for (int j = 1; j <= i; j++) cout << ch << " "; cout << endl; ch++; }
3. This code adds numbers from 1 to 5. sum = 0; for (int i = 1; i < 5; i++) { sum += i; }	4. This program prints a triangle. int n = 4; for (int i = 1; i <= n; i++) { for (int j = 1; j <= n; j++) cout << "* "; cout << endl; }

2/4

5. Which looping process is best used when the number of iterations is known?

A. While loop   B. For loop   C. Do while loop

D. all looping processes require that the iterations be known

6. Every user-defined function should be declared in the program ——

A. after it is used   B. before it is used

C. at the time of its invocation   D. both a and b

7. The output of the following code:

```
For(int j=1;j<10;j+=2)
   { cout<<j;   j=1;}
```

A. Nothing   B. infinity loop   C. 13579   D. 12345678910

8. The valid name for function is:

A. Function name   B. Find max   C. _functionname   D. 1Print

9.
```
int size=10;  int arr[size],arr2;
read(arr,size);  read(arr2,size);
```

A. True   B. False

Q2: Answer the following (answer 2 only, number one is necessary to answer)   6 pts [2x3]

```
void func (int& A, int &B , int C)
{  do{
     A=C;
     B=A+B;
     C *= 2;
     }while (C<=8)
  return C; }
void main(){
  int A=2, B=3, C = 1;
  func(A,B,C);
  char arr (3)={A, B,C};
    cout<< arr(0)<<"\t"<<arr(1)<<"\t"<<arr(2);}
```

1. Correct the above code.
2. Give the output after the correction.
3. Rewrite the code in point 2 using **for** loop instead of **do..while** loop.

Final Exam of Programming II
University of Basrah
College of Computer Science and Information Technology
Department of Computer Science
2022-2023/ Second Semester

Exam Duration: 3 Hours | First Stage/ First Attempt | Date: 4 /6/2023

N.B. (1) The maximum mark for this paper is 38(8+6+12+12)
(2) The marks for each question are shown on the right side of the question.
(3) Answers to questions should be grouped and written together, i.e., all responses to sub-questions of individual questions like Q1, Q2, Q3, etc., should be answered one after the other.
(4) This exam has Two pages. Make sure you have all the pages before you begin.
(5) Please read each question carefully before you start to answer it.

Q1: Choose the correct answer (Answer 8 only) 8 pts [1x8]

1. **The output is:**

```
{int k=20;
 while(k>10)
   {
   cout<<k<<"  ";
   k-=2;
   }
}
```

A. 20 18 16 14 12 10
B. 20 18 16 14 12
B. Nothing
D. infinity loop

2. ______ **is correctly declares an array?**

A. int array(10); B. bool array[10]; C. array{10}; D. array array[10]

3. **The index of the last element of an array with size 9 is ______**

A. 9 B. Programmer-defined C. 0 D. 8

4. **Array is defined as:**

A. An array is a series of element
B. An array is a series of elements of the same type in contiguous memory locations
C. An array is a series of elements of the same type placed in non-contiguous memory locations
D. None of the mentioned.

~~without changing the task of the function.~~

Q3: Answer the following (answer Two only): 12 pts [6x2]

A. **Write a C++ program to print the following pattern:**

```
1   10
2   3   50
4   5   6   150
7   8   9   10   340
11  12  13  14   15   650
```

B. **Write a C++ program to find the result of the following series:**

$$S= a.\frac{x^{2!}}{y+1}+\frac{x^{4!}}{y+2}-a.\frac{x^{6!}}{2y+3}-\frac{x^{8!}}{3y+4}+a.\frac{x^{10!}}{5y+5}+\frac{x^{12!}}{8y+6}-a.\frac{x^{14!}}{13y+7}-\ldots \text{ N term}$$

C. **Write a C++ program to define an array with 11 integer elements, and do the following:**

1. Extract the middle element from the array.
2. Replace each negative element in the array with the value in point 1.
3. Print the array after the replacement process.

Q4: Write a program in C++ to declare 2 array 1st to store ID of movie and the other to store the number of stars)evaluation) that are obtained by the user for the movie. where the program ask user to input id of the movie and then ask to input number of stars then store in stars array, where the stars array contain cumulative number of stars associated with a specific movie. (12 Marks)

Id
101 (Saw)
103 (Scream)
105 (Full)

END OF TEST

Examiner	Head of the Department	The Dean
Lec.Dr. Shatha Falih	Asst. Prof. Dr. Salah F. Salah	Asst. Prof. Dr. Salma Abdulbaki Mahmood
Lec. Huda Adel		
Ass.Lec Qunoot N. Alsahi		

س4: أكتب برنامج بلغة ++C للإعلان عن مصفوفتين، الأولى لتخزين معرف الفيلم (اسناد) والأخرى لتخزين عدد النجوم التي حصل عليها الفلم من قبل المستخدم أو المشاهد. حيث يطلب البرنامج من المستخدم إدخال معرف الفيلم ثم يطلب إدخال عدد النجوم ثم تخزينها في مصفوفة النجوم (يجب أن تكون عدد النجوم محصورة بين 1-5)، حيث تحتوي مصفوفة النجوم على عدد تراكمي من النجوم خاصة بكل فلم.

علما أن المعرفات الخاصة بالافلام هي كالتالي:

Id

101 (Saw)

103 (scream)

105 (Full)